Ashok Budha

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EDUCATION

IOE Thapathali Campus **Thapathali, Kathmandu**
Bachelors in Computer Engineering *Graduation Date: August 2023*
Percentage Scored: 73.49%

- **Courses Relevant:** Computer Programming, Engineering Mathematics, Basic Electrical Engineering, Basic Electronics Engineering, Object Oriented Programming, Theory of Computation, Applied Mathematics, Object Oriented Analysis and Design, Numerical Methods, Computer Graphics, Computer Network, Artificial Intelligence, Data Mining

Sainik Aawasiya Mahavidyalaya **Sallaghari, Bhaktapur**
+2 Science *Graduation Date: May 2018*

Fusemachines **Baneshwor, Kathmandu**
Microdegree in Artificial Intelligence *Course Date: Feb 2023-August 2023*

- Awarded scholarship to attend Fusemachines AI Fellowship 2023.
- This Program offers Fusemachines's own Microdegree programs, with courses holding direct relevance to theoretical and practical aspects of Artificial Intelligence.

RESEARCH PROJECTS

Engineering Major Project: "Automated Multi-Model Pipeline for Generating Football Game Summaries"

- Multi-modal pipeline to generate football game summaries from game audio, metadata and news information.
- Used ROGUE metric as evaluation metric and achieved results comparable to the state-of-the-art models.
- B. Khanal, A. Budha, A. Chapain, K. Shrestha and U. K. Ghimire, "Automated Multi-Modal Pipeline for Generating Football Game Summaries". TechRxiv, 10-Jul-2023, doi:10.36227/techrxiv.23607285.v2.

Engineering Minor Project: "Pneumothorax Segmentation from Chest X-rays using Deep Convolution Networks"

- Pneumothorax segmentation model using deep convolutional networks and use of transfer learning.
- This approach makes early diagnosis of Pneumothorax convenient and shows promising result in use of AI in medical imaging.

Image Captioning using VGG16 and transformer

- Designed and implemented a deep learning pipeline for image captioning, combining VGG16 for image captioning and transformer-based architecture for text generation.

PUBLICATIONS

Automated Multi-Model Pipeline for Generating Football Game Summaries

- Presented the project in **ICT-CEEL (International Conference on Technologies for Computer, Electrical, Electronics & Communication)** conference, 2023 organized in Khwopa Engineering Campus, Bhaktapur.
- Citation: B. Khanal, A. Budha, A. Chapain, K. Shrestha, U. K. Ghimire. "**Automated Multi-Modal Pipeline for Generating Football Game Summaries**." TechRxiv, 10-Jul-2023. doi:10.36227/techrxiv.23607285.v2.

WORK EXPERIENCE

National College of Management and Technical Sciences (NCMT) **Samakhushi, Kathmandu**
Lecturer *November 2025 – Present*

- Deliver undergraduate lectures on Artificial Intelligence for B.Sc. Computer Science and B.Sc. Information Technology programs.
- Design and supervise laboratory sessions and practical assignments to reinforce theoretical concepts.
- Provide academic guidance and mentorship to students pursuing AI-related projects and research.

LIS Nepal Pvt. Ltd. **Manbhawan, Lalitpur**
Software Engineer I *January 2025 – May 2025*

- Participated in the research and development of a natural language to SQL system, beginning with a proof of concept and evolving into a fully developed, deployable solution.
- Developed and maintained NLP parsing modules to extract meaningful information from unstructured text using techniques such as dependency parsing, POS tagging, and NER.
- Designed and optimized domain-specific embeddings for improved metadata handling and industry-specific context capture.
- Built and integrated intent detection and clarification modules to enhance query understanding in conversational systems.
- Developed AI-powered applications using LangChain and LangGraph to interface with large language models (LLMs).
- Designed and maintained scalable backend APIs using Python, Django REST Framework, and FastAPI.
- Conducted research and proof-of-concept (POC) development for machine learning applications, contributing to system design and architecture.
- Collaborated with cross-functional teams, conducted requirement analysis, and performed peer code reviews to ensure robust, production-ready software.

Associate Software Engineer *October 2023 – January 2025*

- Worked on the development of speech-to-text systems using libraries and services such as Vosk, Sphinx, Whisper, Azure Speech-to-Text, and Google Speech-to-Text.
- Created a Retrieval-Augmented Generation (RAG) system for natural language interactions over structured knowledge bases, improving data accessibility and user engagement.
- Built evaluation pipelines for RAG systems and implemented state-of-the-art metrics to assess and enhance generated outputs.
- Assisted in backend development tasks and gained experience with NLP workflows, model deployment, and AI system integration.

Elytra Solutions Pvt. Ltd. **Baneshwor, Kathmandu**
Software Engineer I *May 2023 – October 2023*

- Developed robust and scalable backend systems using Django and Django REST Framework, with an emphasis on designing clean, modular RESTful APIs.
- Integrated asynchronous task queues using Celery and Redis to enable background processing and system decoupling.
- Collaborated closely with frontend developers to define API contracts and deliver a seamless full-stack development experience.
- Implemented secure authentication mechanisms (JWT/OAuth2) and integrated external APIs.

VOLUNTEERING EXPERIENCE

ECAST, Thapathali **Thapathali, Kathmandu**
Python Django Tutor *October 2021 - November 2021*

- Conducted Python Django workshop in Thapathali Campus for ECAST (Electronics and Computer Club Amidst Students of Thapathali): a department-based club for Department of Electronics and Computer Engineering, Thapathali Campus.
- Responsible for teaching the basics of Python, HTML, CSS, Git, GitHub and Django.

Program Coordinator *May 2020 – October 2022*

- Worked as a program coordinator in ECAST Thapathali.
- Coordinated in a team to organize several coding related workshops and organized competitive programming competition "Coding Universe".

ACHIEVEMENTS

LOCUS 2022 Docsumo Datarush Winner (2021)

- Winner of ML competition organized by LOCUS, IOE Pulchowk Campus.
- The task involved classifying abstracts of research papers into several categories.

Semester Topper 8th Semester

- Semester Topper (8th Semester) of Computer Engineering, IOE Thapathali Campus with percentage: 82.53%.

REFERENCES

- **Er. Umesh Kanta Ghimire** | Head of Computer Engineering Department, Thapathali Campus, Institute of Engineering, Tribhuvan University | E-mail: *ukg@tcioe.edu.np*
- **Asst. Professor Dinesh Baniya Kshatri** | Course Instructor, Data Mining and Microprocessor, Department of Computer Engineering, Thapathali Campus, Institute of Engineering, Tribhuvan University | Email: *dbaniya@tcioe.edu.np*
- **Assoc. Professor Shanta Maharjan** | Department Head of (Electronics Management Information System) EMIS Unit, Thapathali Campus, Institute of Engineering, Tribhuvan University | Email: *shantarup@tcioe.edu.np*

LANGUAGES

ENGLISH
- **Full Professional Proficiency**

NEPALI
- **Full Professional Proficiency**